25COP928
Professionalism, Ethics and Cyber Security

Information governance

Dr Parisa Derakhshan

Email p.derakhshan@lboro.ac.uk

TikTok data access

- **31 January 2024** – TikTok CEO Shou Chew addressed a US congressional hearing regarding data handling practices.

- This statement was made in response to concerns about **data access, national security, and user privacy**.

Read more: https://wapo.st/3Om75de.

Leveraging NHS Data for AI Development

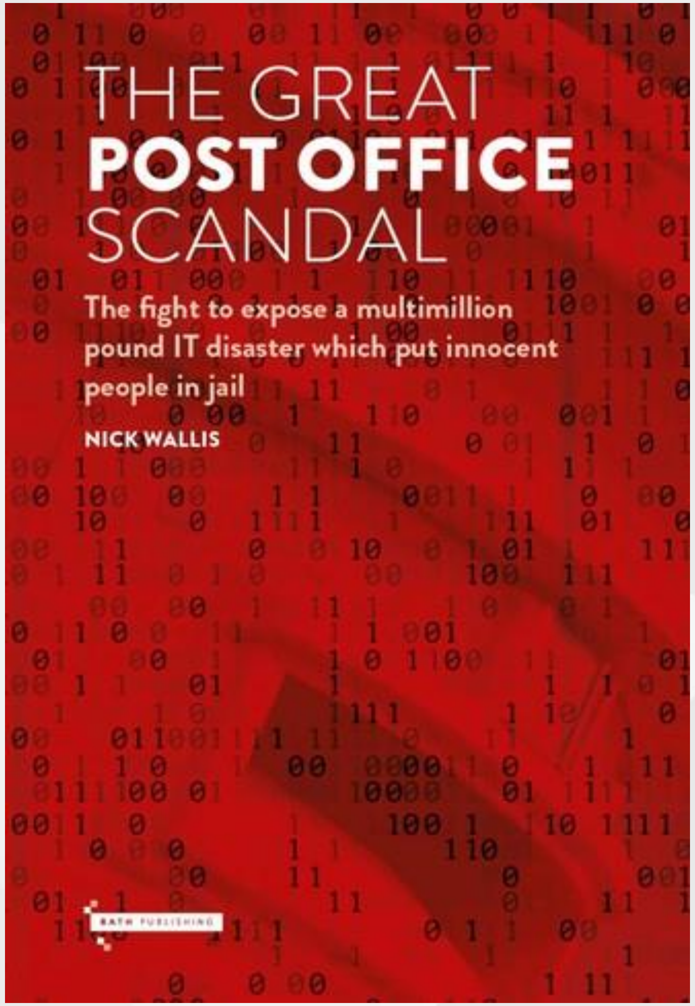
- The UK government is considering allowing **private companies to use anonymised NHS data** to support **AI development in healthcare**.
- Prime Minister **Keir Starmer** has announced plans to **open selected government datasets**, including anonymised patient records, to develop **new treatments and diagnostic tools**.
- While this may drive innovation, it raises concerns about **data privacy, public trust, and ethical use of health data**.

https://www.theguardian.com/society/2025/jan/13/ministers-mull-allowing-private-firms-to-make-profit-from-nhs-data-in-ai-push

Post Office Horizon IT Scandal

- **Faulty accounting software** was used as evidence, leading to the **wrongful prosecution of postmasters** for theft and fraud.

- A later inquiry revealed **systemic failures in data accuracy, management,** and **oversight**, highlighting the importance of strong data governance in preventing miscarriages of justice.



Outlines of the lecture

- What is information governance ?
- Data types and volume
- What Constitutes Information Governance?
- Key areas of information governance

Part I

Organisational Data and Information Governance

What is Information Governance(IG)?

- **IG is about taking control of all organisational information**

- Organisations must get **Information Governance right** to:
  - Comply with **increasing regulatory requirements** across industries
  - Maximise the **value of information** to support innovation, sales, and customer service

- **Poor IG** can lead to **significant financial, legal, and reputational costs**

The volume of data that exists is growing exponentially !!!

Data types and volume

Types of data organisations manage:

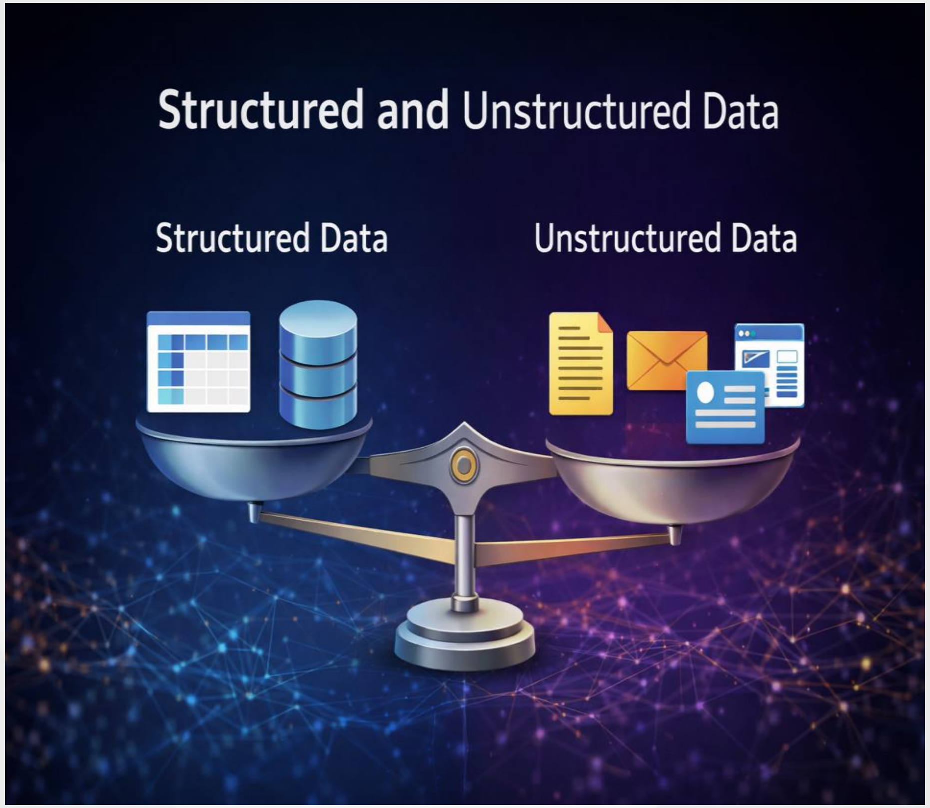
- Structured and Unstructured Data
- Electronic and Paper Documents
- Internal and External Data
- Digital, Video and Audio Data

Structured vs unstructured data

- **Structured data**
  - Stored in **fixed, predefined fields**
  - Found in **databases, ERP systems, eCommerce,and OLTP applications**

- **Unstructured data**
  - No predefined format
  - Makes up the **majority of organisational data**
  - Includes **documents, emails, and presentations**

Data catalogues are used to describe and manage both structured and unstructured data.



Electronic and Paper Documents

- **Electronic and Paper Records:** Organisations manage both electronic and paper-based information.
- **Secure Storage:** Paper records must be stored securely, just like electronic data.
- **Accessibility and Retrieval:** Information should be easy to retrieve when needed, regardless of format.

Internal and External Data

- **Bring Your Own Device (BYOD):** Employees use personal devices at work. Corporate information may be stored outside organisational systems, making it harder to control.
- **Social Media and Online Communication:** Employees interact with customers, suppliers, and partners through social media and online platforms.
- **Governance and Risk:** Information shared through internal and external channels can affect the organisation. It must be properly governed to reduce data protection and information security risks.

Digital, Video and Audio Data

Growing data volume

Organisations face rapidly increasing data from more devices, communication tools, and formats.

- **Video data:** Meetings, webinars, and online content.
- **Audio data:** Recorded calls and voice data (e.g., finance, healthcare).

Managing this data

To make good use of data, organisations need a clear strategy for how information is collected, used, stored, accessed, and securely deleted.

Information Governance guides how data is used to create business-value information.

Types of data

Public

E.g. Policy details

Confidential

E.g. Bank customer details, test cases,

Restricted or private data

Extra sensitive data, e.g: password, system/network design

Successful Information Governance

- **Control and Value**
  - Manage information across its entire lifecycle (create → use → archive → delete)
  - Understand what data exists, where it is, its format, quality, and value.
- **Management and Compliance**
  - Define and communicate clear policies and procedures
  - Ensure information is stored correctly and used appropriately
  - Archive and defensibly delete information when no longer needed
- **Access and Security**
  - Make information available when required
  - Maintain secure access, confidentiality, and protection
- **People and Improvement**
  - Support compliance through training and communication
  - Adapt to new data formats and communication channels
  - Embed continuous improvement

Information Governance

Definition:

Information Governance (IG) is the effective use and management of an organisation's information assets to maximise value while minimising information-related risk.

It brings together the rules, regulations, legislation, standards, and policies that govern how information is created, shared, stored, and used within the organisation.

Part II

What Constitutes Information Governance?

What Constitutes Information Governance?

Information Governance (IG)

- Defines policies and processes for managing organisational information
- Maximises value while reducing information risk
- Applies to all information, regardless of format or location
- Ongoing, organisation-wide programme

So, what makes up an Information Governance programme?

Components of an Information Governance Programme

- Council/committee
- Defining valuable data
- Reaching the entire organisation
- Legal department
- Risk management
- Compliance
- Records management
- Information technology

Information Governance council

Leadership and Oversight

An Information Governance programme should be led by senior, cross-functional leaders through a **dedicated IG Committee or Council.**

Inclusive Representation

The Council should include **representatives from business areas** that are directly affected at different stages of the programme.

Rotating Membership

Membership of the Council should **rotate over time** to encourage broader involvement and fresh perspectives.

Defining the Value of Business Information

Valuable	No Value
Unique	Duplicated
Accessible	Inaccessible
Current	Obsolete
Up-to-date	Incomplete
Categorised/searchable	Uncategorised
Productive for user	Useless to user

Reaching the entire organisation

Organisation-wide involvement

Information Governance affects every department, so all areas must be involved.

Shared ownership

Each business area helps shape the policies and procedures that apply to them.

RIM (Records and Information Management) ensures organisational information is properly managed throughout its lifecycle.

Legal Department

Policy & Privacy

- Define rules for using email, social media, and mobile devices
- Ensure privacy and legal compliance

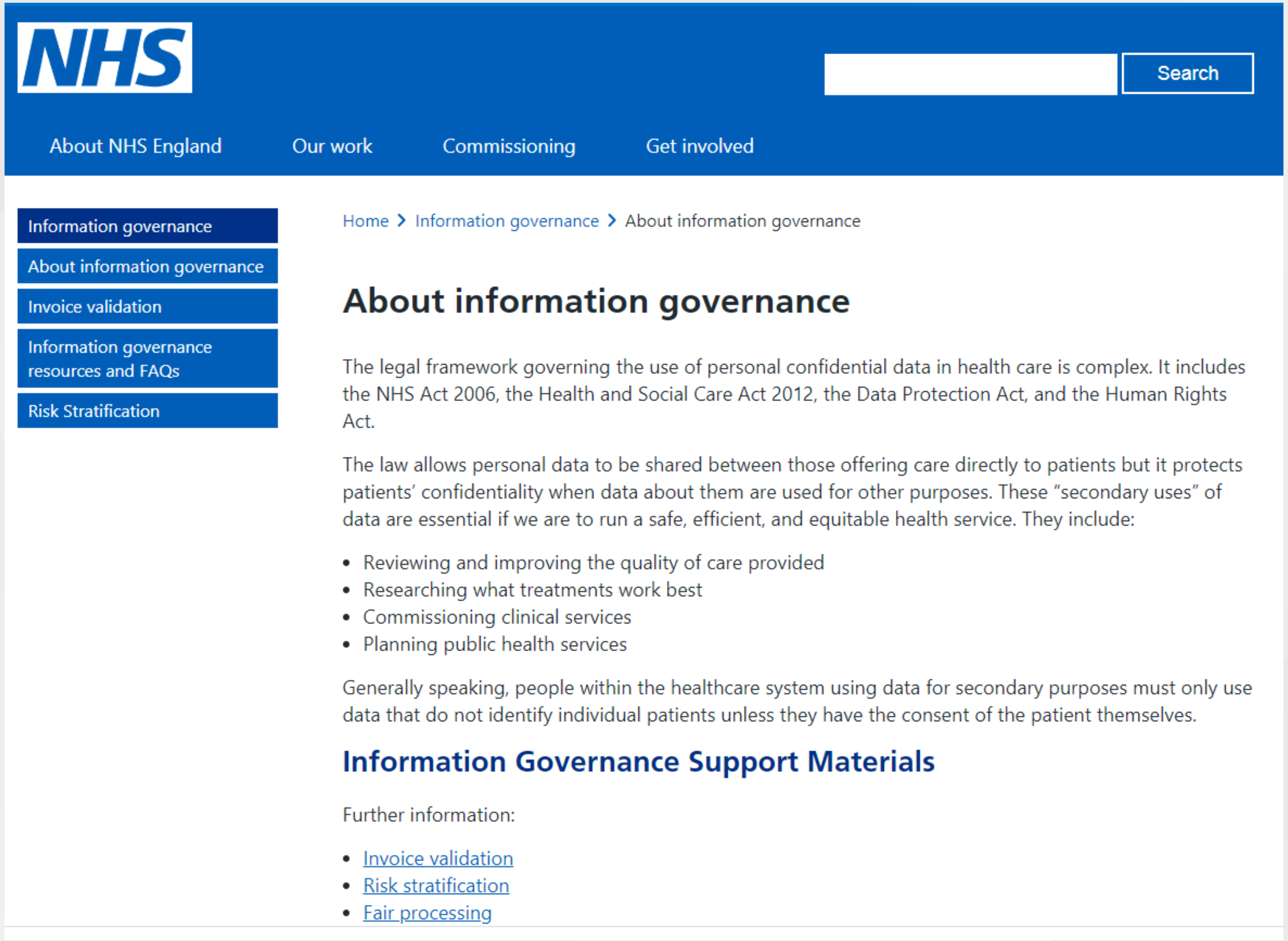
Defensible Disposal

- Approve when information can be safely deleted
- Manage and communicate legal holds to business units

eDiscovery

- Set rules for where information is stored
- Guide how information is accessed and presented when required

Legal Department example of NHS IG


NHS

Search

About NHS England | Our work | Commissioning | Get involved

- Information governance
- About information governance
- Invoice validation
- Information governance resources and FAQs
- Risk Stratification

Home > Information governance > About information governance

About information governance

The legal framework governing the use of personal confidential data in health care is complex. It includes the NHS Act 2006, the Health and Social Care Act 2012, the Data Protection Act, and the Human Rights Act.

The law allows personal data to be shared between those offering care directly to patients but it protects patients' confidentiality when data about them are used for other purposes. These "secondary uses" of data are essential if we are to run a safe, efficient, and equitable health service. They include:

- Reviewing and improving the quality of care provided
- Researching what treatments work best
- Commissioning clinical services
- Planning public health services

Generally speaking, people within the healthcare system using data for secondary purposes must only use data that do not identify individual patients unless they have the consent of the patient themselves.

Information Governance Support Materials

Further information:

- Invoice validation
- Risk stratification
- Fair processing

About information governance

Risk Management

- **Works with Legal** to reduce information-related risks, including litigation, regulatory compliance, and reputational damage
- **Collaborates with IT** on disaster recovery and business continuity planning
- **Maintains oversight of information** by understanding where data is stored, how it is protected, and how it is securely destroyed

Compliance

- Defines how information is **stored, accessed, and controlled**
- Establishes **internal controls and monitoring** for information use
- Manages **audits** and responds to **regulators and external auditors**

Records Management

- **Organisation and storage:** Focuses on how paper and electronic documents are organised, stored, and managed within the organisation.
- **Policies and procedures:** Defines how new types of information (e.g., BYOD, social media, cloud services) are captured and managed.
- **Information lifecycle:** Works across the lifecycle, from creation and use to storage and disposal.
- **Compliance:** Works closely with Compliance to ensure information is handled correctly, legally, and in line with organisational policies.

Information Technology

Data Management: Handles the growing amount of organisational data.

Infrastructure & Storage: Optimises systems and cloud storage to support data use.

System Rationalisation: Removes outdated and duplicate systems.

Supports Governance: Ensures secure access and protection of data throughout its lifecycle.

What constitutes Information Governance?

The Questions

Any Information Governance programme should answer the following questions:

- What information does the organisation have?
- Why is the information needed?
- Who should access and use the information?
- How, when and where can they use the information?
- What can they do with the information?
- Where is the information stored?
- How can the information be share with employees, partners and suppliers?

Council Members

An Information Governance Council should include job roles such as:

- Legal Officer
- Discovery/Litigation Officer
- Records Manager
- CIO
- Compliance Officer
- Impacted Line of Business Managers
- Chief Data Officer
- IT Security Officer

Benefits Information Governance

- Helps organisations **understand what information they have** and **why it is valuable**
- Ensures information is accessed **securely**, by the **right people**, at the **right time**
- Supports **better decision-making**, **business analytics**, and **collaboration**
- Reduces **risk**, **cost**, and **duplication of information**

Challenges of Information Governance

- Organisations of all sizes face a **rapid growth of information** across systems, platforms, and locations
- Increasing **governmental, financial, and industry regulations** place pressure on how information is managed
- More frequent **legal and litigation requests** require organisations to find, protect, and produce information quickly

Part III

Big Data

Big Data

- **Big data** refers to the rapid growth and availability of **large volumes of structured and unstructured data**.

Big Data: Importance and Risks

- This data is often **too large or complex** for traditional data management tools
- Organisations see big data as a **key business asset**, supporting insight, efficiency, and future growth
- However, big data initiatives can lead to **over-collection of information**, increasing **cost, risk, and complexity**

Information Governance and Big Data

- **The Challenge:** Rethinking information management in the big data era
- **The Risk:** Keeping all data is costly and increases legal and compliance risks.
- **The Solution:** Information Governance ensures only valuable information is retained.
- **The Scope:** Applies to structured and unstructured data, including new channels.

Information Governance for Big Data: key questions to consider

- How information is to be stored, identified, collected and reviewed ?
- Which communication channels are used within the organisation and how information in those channels is created and communicated?
- What is business value of data (structured and unstructured) is to the organisation?
- How your Big data programmes operate in conjunction with regulatory compliance and eDiscovery responsibilities
- When information can be removed from corporate systems and the process for disposal

Key areas of Information Governance for Big Data

Retention management
- Data lifecycle,
- Data retention policy

Records management
- policy and practices applied according to the business value

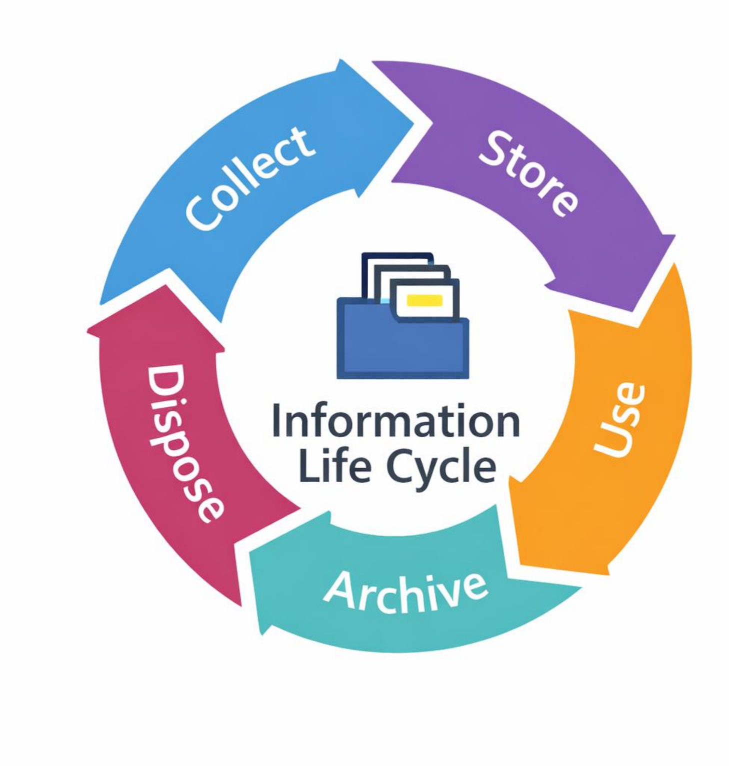
Defensible disposal
- Storage growth and costs,
- Regulatory requirements

Information storage
- Reduce the amount of unnecessary information retained

Social media
- Social media communications channels must be managed in companies

Data retention management

- Ensures information is **not kept forever**
- Applies a **defined lifecycle** to information (this may vary across departments)
- A **retention policy** defines:
  - How long information must be kept
  - What happens when the retention period ends (e.g. archive or delete)

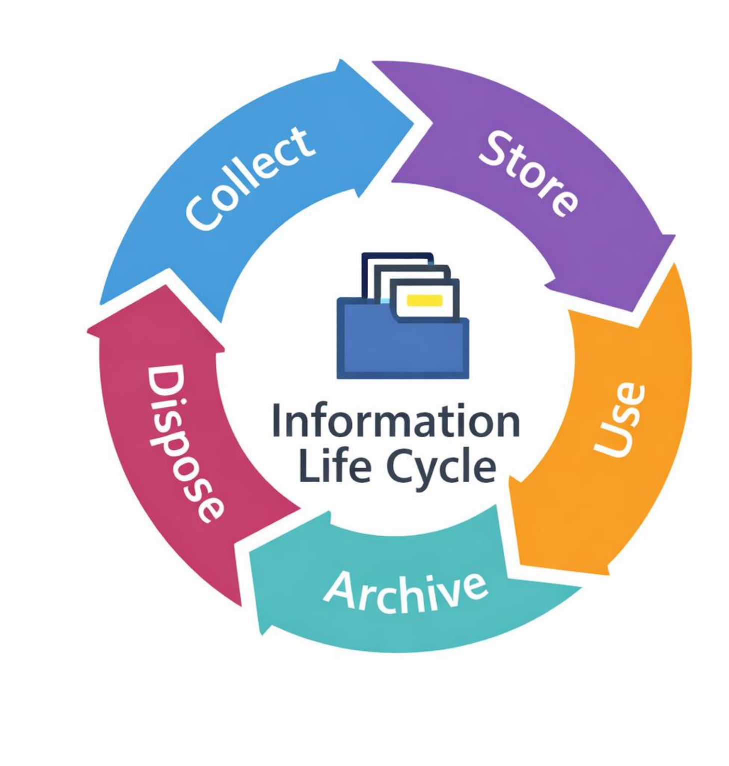


Records management

- Defines policies based on the **business value** of information.
- Uses **automatic categorisation** to improve access and control.
- Manages records across their **full lifecycle**, from creation to deletion.
- A core part of **Information Governance** for unstructured information.

Information Defensible Disposal

- **Controls storage growth and costs**
- **Meets legal and regulatory requirements**
- **Supports Big Data by improving data quality**
- **Removes irrelevant and duplicate information**
- **Especially important for unstructured data** (emails, documents)

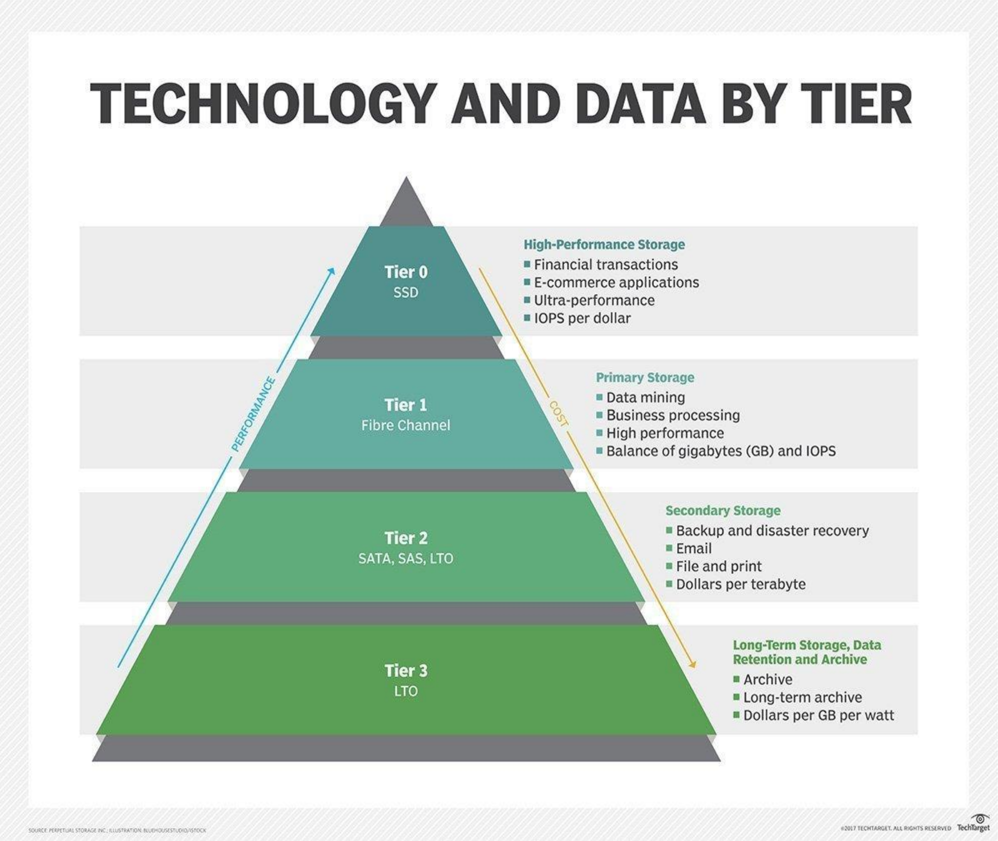


Information storage

Storing everything **increases costs quickly** as data becomes harder and more expensive to manage.

How Information Governance helps:

- **Reduces pressure on IT systems** by removing unnecessary or low-value information.
- **Automates movement between storage tiers** e.g. from **Tier 1** (high-performance) to **Tier 2** (lower-cost) storage.
- **Identifies obsolete systems and data** and supports modern techniques such as **virtualisation**.



[Definition information lifecycle management]

Social media

- **Business information now comes from many channels**, not just internal systems.
- **Social media platforms** (e.g. Facebook, X/Twitter, LinkedIn) contain valuable insights on **customers and market trends**.
- **Instant messaging and social media must be included** in any Information Governance programme.
- **A unified Information Governance platform** can collect this data and feed it into **analytics and Big Data systems**.